

LESSON 3: FORCES AND CENTRES OF PRESSURE OF PLANA AND NON-PLANA SURFACES

Definitions:

Planar Surface: Is defined as a surface that is a plane, where a plane is a flat 2-dimensional surface that is straight in two directions.

Non-Planar Surface: is one that is not flat or level in all directions, meaning its vertices do not all lie on the same plane

Total Pressure: The force exerted by a static fluid on a surface either plane or curved when the fluid comes in contact with the surfaces. The force acts normal to the surface.

Centre of pressure: The point of application of the total pressure surface. Submerged surfaces may take the following forms:

- i. Vertical plane surface
- ii. Horizontal plane surface
- iii. Inclined plane surface
- iv. Curved surface

Consider a vertical plane surface of arbitrary shape immersed in a liquid as shown in fig. 3.1

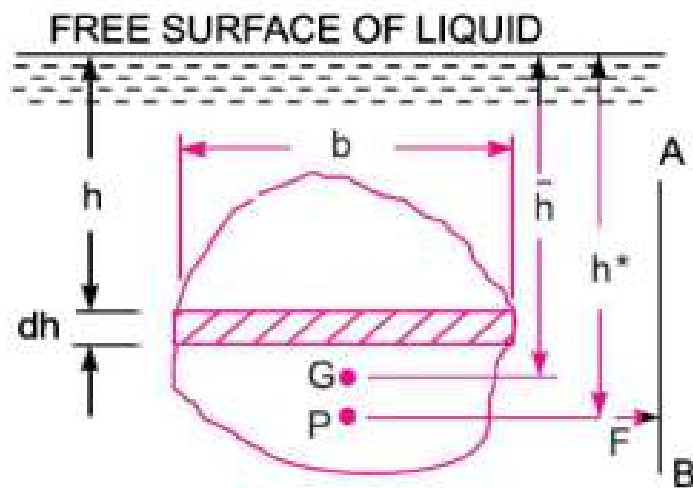


Fig. 3.1

Let A = Total area of the surface

$\bar{h}$ = Distance of C.G. of the area from free surface of liquid

G = Centre of gravity of plane surface

P = Centre of pressure

h^* = Distance of centre of pressure from free surface of liquid.

a) Total Pressure on the surface can be determined by dividing the entire surface into a number of small parallel strips.

The force on each small strip is then calculated and while the total pressure force on the whole area calculated by integrating the force on small strip.

Consider that the thickness of the strip is dh and width b at a depth of h from free surface of the liquid as shown in fig. 3.1

Pressure intensity on the strip, $p = \rho gh$

Area of the strip, $dA = b \times dh$

Total pressure force on strip, $dF = p \times \text{Area}$
 $= \rho gh \times b \times dh$

$\therefore$ Total pressure force on the whole surface,

$$F = \int dF = \int \rho gh \times b \times dh = \rho g \int b \times h \times dh$$

But $\int b \times h \times dh = \int h \times dA$

= Moment of surface area about the free surface of liquid

= Area of surface $\times$ Distance of C.G. from free surface

$$= A \times \bar{h}$$

$$\therefore F = \rho g A \bar{h} \quad \text{eq (i)}$$

b) Centre of Pressure (h^*): This is calculated using the “**Principle of Moments**” which states that the moment of resultant force about an axis is equal to the sum of moments of the components about the same axis

The resultant force F is acting at P , at a distance h^* from free surface of the liquid as shown in Fig. 3.1.

Force about free surface of the liquid = $F \times h^*$

eq (ii)

Moment of force dF , acting on a strip about free surface of liquid

$$\begin{aligned} &= dF \times h & \{ \because dF = \rho g h \times b \times dh \} \\ &= \rho g h \times b \times dh \times h \end{aligned}$$

Sum of moments of all such forces about free surface of liquid

$$\begin{aligned} &= \int \rho g h \times b \times dh \times h = \rho g \int b \times h \times h dh \\ &= \rho g \int b h^2 dh = \rho g \int h^2 dA & (\because b dh = dA) \end{aligned}$$

But

$$\int h^2 dA = \int b h^2 dh$$

= Moment of Inertia of the surface about free surface of liquid

$$= I_0$$

$\therefore$ Sum of moments about free surface

$$= \rho g I_0$$

eq (iii)

Equating Eq. (ii) and (iii), we get

$$F \times h^* = \rho g I_0$$

But $F = \rho g A \bar{h}$

$$\therefore \rho g A \bar{h} \times h^* = \rho g I_0$$

or
$$h^* = \frac{\rho g I_0}{\rho g A \bar{h}} = \frac{I_0}{A \bar{h}} \quad \text{eq (iv)}$$

By the theorem of parallel axis, we have

$$I_0 = I_G + A \times \bar{h}^2$$

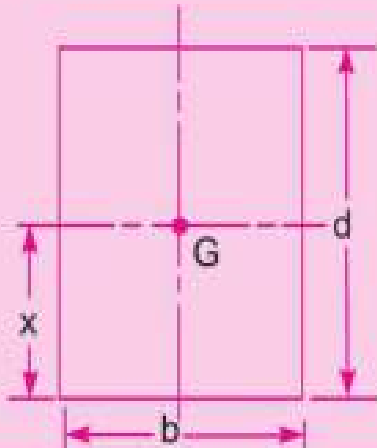
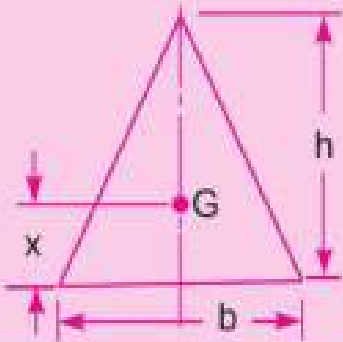
Where I_G = Moment of inertia of area about an axis passing through the centre of gravity of the area and parallel to the free surface of liquid.

Substituting I_0 in equation (iv), we get

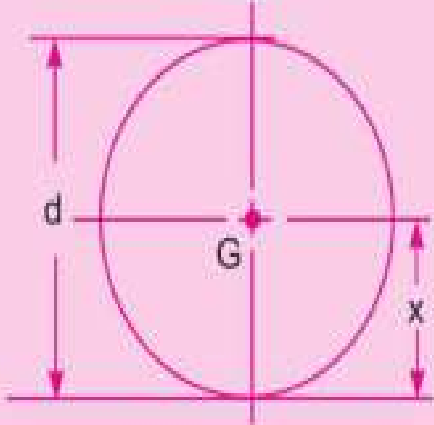
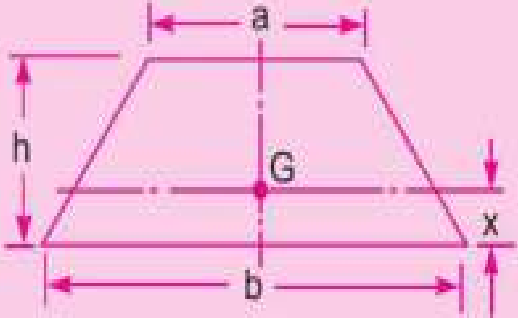
$$h^* = \frac{I_G + A \bar{h}^2}{A \bar{h}} = \frac{I_G}{A \bar{h}} + \bar{h} \quad \text{eq (v)}$$

In Eq. (v), $\bar{h}$ is the distance of C.o.G of the area of the vertical surface from the free surface of the liquid.

Table 3.1 The moments of inertia and other geometric properties of some important plane surfaces

<i>Plane surface</i>	<i>C.G. from the base</i>	<i>Area</i>	<i>Moment of inertia about an axis passing through C.G. and parallel to base (I_G)</i>	<i>Moment of inertia about base (I_0)</i>
1. Rectangle 	$x = \frac{d}{2}$	bd	$\frac{bd^3}{12}$	$\frac{bd^3}{3}$
2. Triangle 	$x = \frac{h}{3}$	$\frac{bh}{2}$	$\frac{bh^3}{36}$	$\frac{bh^3}{12}$

Contd...

<i>Plane surface</i>	<i>C.G. from the base</i>	<i>Area</i>	<i>Moment of inertia about an axis passing through C.G. and parallel to base (I_G)</i>	<i>Moment of inertia about base (I_o)</i>
3. Circle  The diagram shows a circle with a vertical diameter labeled d. The center is marked with a dot and labeled G. A horizontal line segment from the base to the center is labeled x.	$x = \frac{d}{2}$	$\frac{\pi d^2}{4}$	$\frac{\pi d^4}{64}$	—
4. Trapezium  The diagram shows a trapezium with a top horizontal edge of length a and a bottom horizontal edge of length b. The height is labeled h. A horizontal dashed line passes through the center of gravity G. The distance from the bottom base to G is labeled x.	$x = \left(\frac{2a + b}{a + b} \right) \frac{h}{3}$	$\frac{(a + b)}{2} \times h$	$\left(\frac{a^2 + 4ab + b^2}{36(a + b)} \right) \times h^3$	—

Worked Example 1:

A rectangular plane surface is 2 m wide and 3 m deep. It lies in vertical plane in water. Determine the total pressure and position of centre of pressure on the surface when its upper edge is horizontal and (a) coincides with water surface, (b) 2.5 m below the free water surface.

Solution. Given :

Width of plane surface, $b = 2 \text{ m}$

Depth of plane surface, $d = 3 \text{ m}$

(a) **Upper edge coincides with water surface (Fig. 3.2).** Total pressure is given by equation as

$$F = \rho g A \bar{h}$$

where $\rho = 1000 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$

$$A = 3 \times 2 = 6 \text{ m}^2, \bar{h} = \frac{1}{2} (3) = 1.5 \text{ m.}$$

$$\therefore F = 1000 \times 9.81 \times 6 \times 1.5 \\ = 88290 \text{ N. Ans.}$$

Depth of centre of pressure is given by equation (3.5) as

$$h^* = \frac{I_G}{A\bar{h}} + \bar{h}$$

where $I_G = \text{M.O.I. about C.G. of the area of surface}$

$$= \frac{bd^3}{12} = \frac{2 \times 3^3}{12} = 4.5 \text{ m}^4$$

$$h^* = \frac{4.5}{6 \times 1.5} + 1.5 = 0.5 + 1.5 = 2.0 \text{ m. Ans.}$$

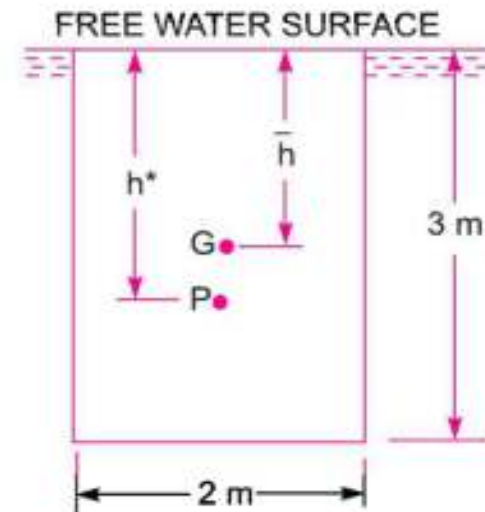


Fig. 3.2

(b) Upper edge is 2.5 m below water surface (Fig. 3.3). Total pressure (F) is given by (eq (i))

$$F = \rho g A \bar{h}$$

where $\bar{h}$ = Distance of C.G. from free surface of water

$$= 2.5 + \frac{3}{2} = 4.0 \text{ m}$$

$\therefore$

$$F = 1000 \times 9.81 \times 6 \times 4.0 \\ = \mathbf{235440 \text{ N. Ans.}}$$

Centre of pressure is given by $h^* = \frac{I_G}{Ah} + \bar{h}$

where $I_G = 4.5$, $A = 6.0$, $\bar{h} = 4.0$

$\therefore$

$$h^* = \frac{4.5}{6.0 \times 4.0} + 4.0$$

$$= 0.1875 + 4.0 = 4.1875 = \mathbf{4.1875 \text{ m. Ans.}}$$

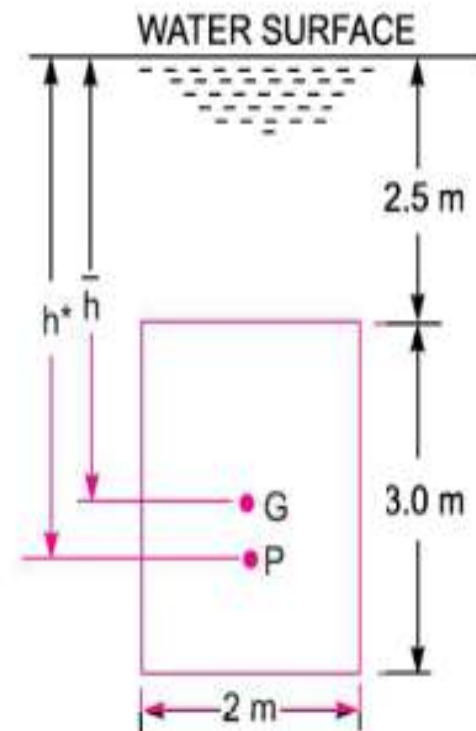


Fig. 3.3

Worked Example 2:

Determine the total pressure on a circular plate of diameter 1.5 m which is placed vertically in water in such a way that the centre of the plate is 3 m below the free surface of water. Find the position of centre of pressure also.

Solution. Given : Dia. of plate, $d = 1.5$ m

$$\therefore \text{Area,} \quad A = \frac{\pi}{4} (1.5)^2 = 1.767 \text{ m}^2$$

$$\bar{h} = 3.0 \text{ m}$$

Total pressure is given by equation (3.1),

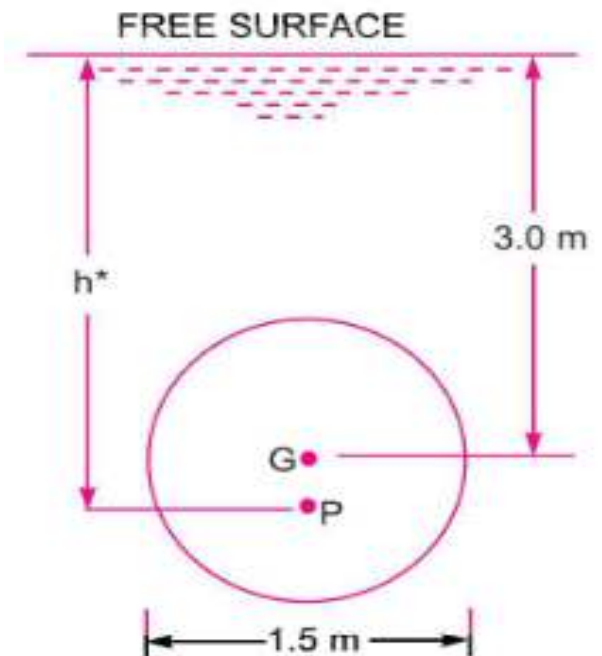
$$\begin{aligned} F &= \rho g A \bar{h} \\ &= 1000 \times 9.81 \times 1.767 \times 3.0 \text{ N} \\ &= \mathbf{52002.81 \text{ N. Ans.}} \end{aligned}$$

Position of centre of pressure (h^*) is given by equation (3.5),

$$h^* = \frac{I_G}{A\bar{h}} + \bar{h}$$

$$\text{where } I_G = \frac{\pi d^4}{64} = \frac{\pi \times 1.5^4}{64} = 0.2485 \text{ m}^4$$

$$\begin{aligned} \therefore h^* &= \frac{0.2485}{1.767 \times 3.0} + 3.0 = 0.0468 + 3.0 \\ &= \mathbf{3.0468 \text{ m. Ans.}} \end{aligned}$$



Worked Example 3:

A trapezoidal channel 2 m wide at the bottom and 1 m deep has side slopes 1:1.

Determine: i) the total pressure and ii) the centre of pressure on the vertical gate closing the channel when it is full of water.

Solution. Given :

Width at bottom $\quad\quad\quad = 2 \text{ m}$

Depth, $\quad\quad\quad d = 1 \text{ m}$

Side slopes $\quad\quad\quad = 1 : 1$

$\therefore$ Top width, $\quad\quad\quad AD = 2 + 1 + 1 = 4 \text{ m}$

Area of rectangle $FBEC$, $\quad\quad\quad A_1 = 2 \times 1 = 2 \text{ m}^2$

Area of two triangles ABF and ECD , $A_2 = \frac{(4-2)}{2} \times 1 = 1 \text{ m}^2$

$\therefore$ Area of trapezoidal $ABCD$, $A = A_1 + A_2 = 2 + 1 = 3 \text{ m}^2$

Depth of C.G. of rectangle $FBEC$ from water surface,

$$\bar{h}_1 = \frac{1}{2} = 0.5 \text{ m}$$

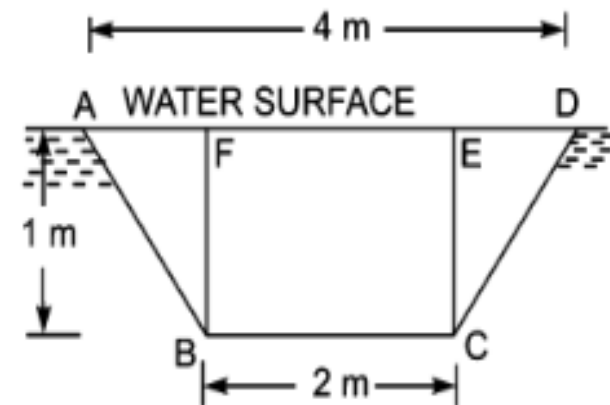


Fig. 3.12

Depth of C.G. of two triangles ABF and ECD from water surface,

$$\bar{h}_2 = \frac{1}{3} \times 1 = \frac{1}{3} \text{ m}$$

$\therefore$ Depth of C.G. of trapezoidal $ABCD$ from free surface of water

$$\bar{h} = \frac{A_1 \times \bar{h}_1 + A_2 \times \bar{h}_2}{(A_1 + A_2)} = \frac{2 \times 0.5 + 1 \times 0.33333}{(2 + 1)} = .44444$$

(i) **Total Pressure (F).** Total pressure F is given by

$$\begin{aligned} F &= \rho g A \bar{h} \\ &= 1000 \times 9.81 \times 3.0 \times 0.44444 = \mathbf{13079.9 \text{ N. Ans.}} \end{aligned}$$

(ii) **Centre of Pressure (h^*).** M.O.I. of rectangle $FBCE$ about its C.G.,

$$I_{G_1} = \frac{bd^3}{12} = \frac{2 \times 1^3}{12} = \frac{1}{6} \text{ m}^4$$

M.O.I. of $FBCE$ about an axis passing through the C.G. of trapezoidal

or $I_{G_1}^* = I_{G_1} + A_1 \times [\text{Distance between C.G. of rectangle and C.G. of trapezoidal}]^2$

$$= \frac{1}{6} + 2 \times [\bar{h}_1 - \bar{h}]^2$$

$$= \frac{1}{6} + 2 \times [0.5 - .4444]^2 = .1666 + .006182 = 0.1727$$

M.O.I. of the two triangles about the C.G. of trapezoidal,

$$I_{G_2}^* = I_{G_1} + A_2 \times [\text{Distance between C.G. of triangles and C.G. of trapezoidal}]^2$$

$$= \frac{1}{18} + 1 \times [\bar{h} - \bar{h}_2]^2 = \frac{1}{18} + 1 \times \left[.4444 - \frac{1}{3} \right]^2$$

$$= \frac{1}{18} + (.1111)^2 = 0.0555 + (.1111)^2$$

$$= .0555 + 0.01234 = 0.06789 \text{ m}^4$$

∴ M.O.I. of the trapezoidal about its C.G.

$$I_G = I_{G_1}^* + I_{G_2}^* = .1727 + .06789 = 0.24059 \text{ m}^4$$

Then centre of pressure (h^*) on the vertical trapezoidal,

$$h^* = \frac{I_G}{A\bar{h}} + \bar{h} = \frac{0.24059}{3 \times .4444} + .4444 = 0.18046 + .4444 = 0.6248$$

$$\approx \mathbf{0.625 \text{ m. Ans.}}$$

Worked Example 4:

A cubical tank has sides of 1.5 m. It contains water for the lower 0.6 m depth. The upper remaining part is filled with oil of specific gravity 0.9. Calculate for one vertical side of the tank: a) Total pressure and b) position of centre of pressure

Solution. Given :

Cubical tank of sides 1.5 m means the dimensions of the tank are 1.5 m $\times$ 1.5 m $\times$ 1.5 m.

Depth of water = 0.6 m

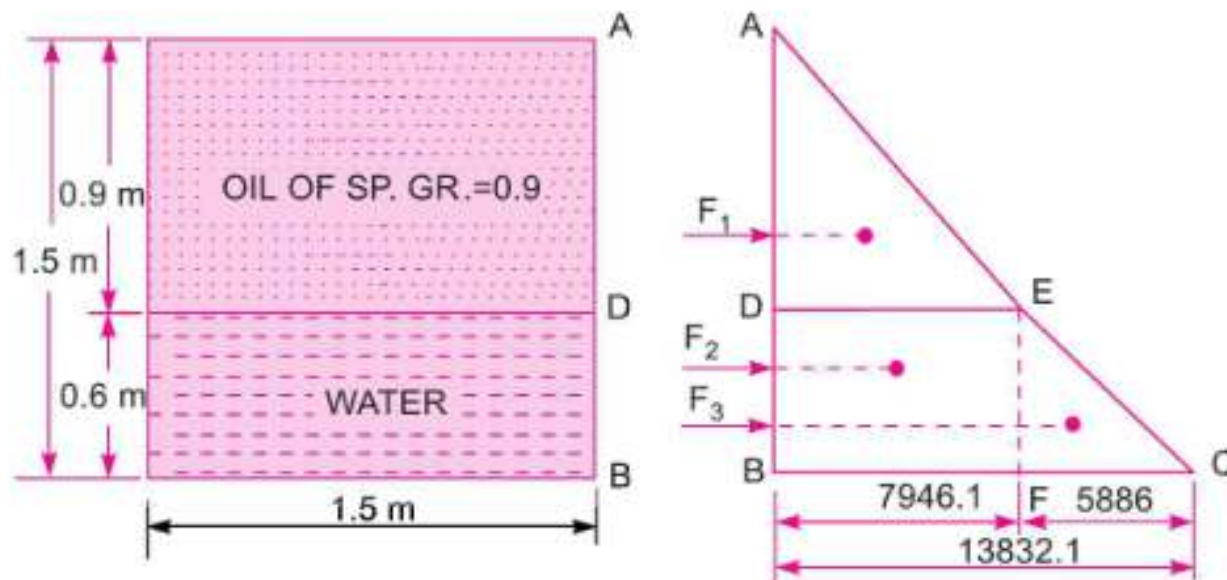
Depth of liquid = 1.5 – 0.6 = 0.9 m

Sp. gr. of liquid = 0.9

Density of liquid, $\rho_1 = 0.9 \times 1000 = 900 \text{ kg/m}^3$

Density of water, $\rho_2 = 1000 \text{ kg/m}^3$

(a) **Total pressure** on one vertical side is calculated by drawing pressure diagram, which is shown in Fig. 3.15.



Intensity of pressure at A , $p_A = 0$

Intensity of pressure at D , $p_D = \rho_1 g \times h = 900 \times 9.81 \times 0.9 = 7946.1 \text{ N/m}^2$

Intensity of pressure at B , $p_B = \rho_1 g h_1 + \rho_2 g h_2 = 900 \times 9.81 \times 0.9 + 1000 \times 9.81 \times 0.6$
 $= 7946.1 + 5886 = 13832.1 \text{ N/m}^2$

Hence in pressure diagram :

$$DE = 7946.1 \text{ N/m}^2, BC = 13832.1 \text{ N/m}^2, FC = 5886 \text{ N/m}^2$$

The pressure diagram is split into triangle ADE , rectangle $BDEF$ and triangle EFC . The total pressure force consists of the following components :

(i) Force $F_1 = \text{Area of triangle } ADE \times \text{Width of tank}$
 $= \left(\frac{1}{2} \times AD \times DE\right) \times 1.5 \quad (\because \text{Width} = 1.5 \text{ m})$
 $= \left(\frac{1}{2} \times 0.9 \times 7946.1\right) \times 1.5 \text{ N} = 5363.6 \text{ N}$

This force will be acting at the C.G. of the triangle ADE , i.e., at a distance of $\frac{2}{3} \times 0.9 = 0.6 \text{ m}$ below A

(ii) Force $F_2 = \text{Area of rectangle } BDEF \times \text{Width of tank}$
 $= (BD \times DE) \times 1.5 = (0.6 \times 7946.1) \times 1.5 = 7151.5$

This force will be acting at the C.G. of the rectangle $BDEF$ i.e., at a distance of $0.9 + \frac{0.6}{2} = 1.2 \text{ m}$

below A .

(iii) Force $F_3 = \text{Area of triangle } EFC \times \text{Width of tank}$
 $= \left(\frac{1}{2} \times EF \times FC\right) \times 1.5 = \left(\frac{1}{2} \times 0.6 \times 5886\right) \times 1.5 = 2648.7 \text{ N}$

This force will be acting at the C.G. of the triangle EFC , *i.e.*, at a distance of $0.9 + \frac{2}{3} \times 0.6 = 1.30$ m

below A .

$\therefore$ Total pressure force on one vertical face of the tank,

$$\begin{aligned} F &= F_1 + F_2 + F_3 \\ &= 5363.6 + 7151.5 + 2648.7 = \mathbf{15163.8 \text{ N. Ans.}} \end{aligned}$$

(b) Position of centre of pressure

Let the total force F is acting at a depth of h^* from the free surface of liquid, *i.e.*, from A .

Taking the moments of all forces about A , we get

$$F \times h^* = F_1 \times 0.6 + F_2 \times 1.2 + F_3 \times 1.3$$

or

$$\begin{aligned} h^* &= \frac{F_1 \times 0.6 + F_2 \times 1.2 + F_3 \times 1.3}{F} \\ &= \frac{5363.6 \times 0.6 + 7151.5 \times 1.2 + 2648.7 \times 1.3}{15163.8} \\ &= \mathbf{1.005 \text{ m from A. Ans.}} \end{aligned}$$

INCLINED PLANE SURFACE SUBMERGED IN LIQUID

Consider a plane surface of arbitrary shape immersed in a liquid in such a way that the plane of the surface makes an angle θ with the free surface of the liquid as shown in Fig. 3.18.

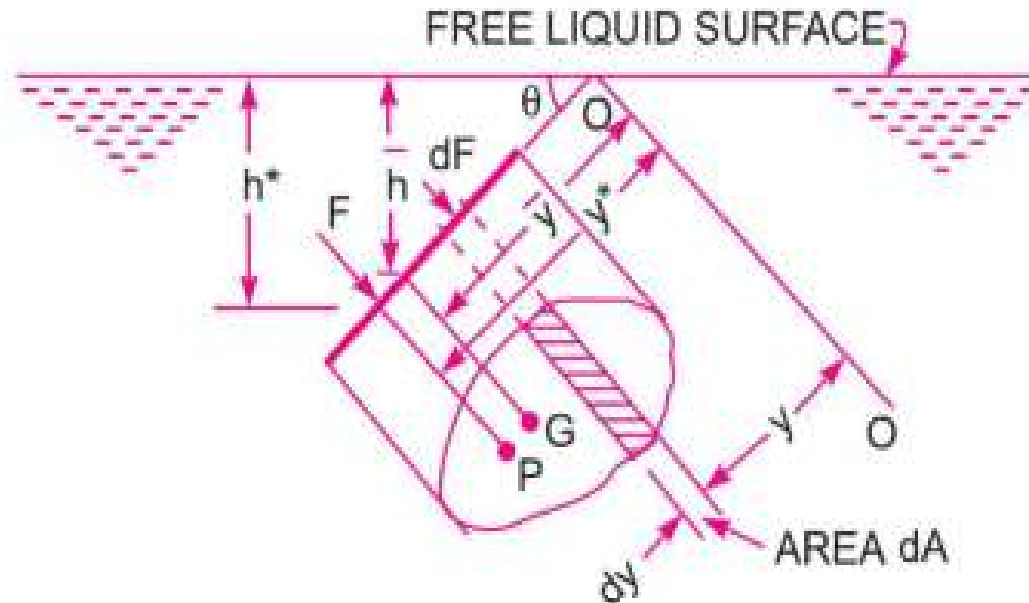


Fig. 3.18 *Inclined immersed surface.*

Let A = Total area of inclined surface

$\bar{h}$ = Depth of C.G. of inclined area from free surface

h^* = Distance of centre of pressure from free surface of liquid

θ = Angle made by the plane of the surface with free liquid surface.

Let the plane of the surface, if produced meet the free liquid surface at O . Then $O-O$ is the axis perpendicular to the plane of the surface.

Let $\bar{y}$ = distance of the C.G. of the inclined surface from $O-O$
 y^* = distance of the centre of pressure from $O-O$.

Consider a small strip of area dA at a depth ' h ' from free surface and at a distance y from the axis $O-O$ as shown in Fig. 3.18.

Pressure intensity on the strip, $p = \rho gh$

$\therefore$ Pressure force, dF , on the strip, $dF = p \times \text{Area of strip} = \rho gh \times dA$

Total pressure force on the whole area, $F = \int dF = \int \rho gh dA$

But from Fig. 3.18, $\frac{h}{y} = \frac{\bar{h}}{\bar{y}} = \frac{h^*}{y^*} = \sin \theta$

$\therefore h = y \sin \theta$

$\therefore F = \int \rho g \times y \times \sin \theta \times dA = \rho g \sin \theta \int y dA$

But $\int y dA = A \bar{y}$

where $\bar{y}$ = Distance of C.G. from axis $O-O$

$$\begin{aligned} F &= \rho g \sin \theta \bar{y} \times A \\ &= \rho g A \bar{h} \end{aligned}$$

eq (vi)

Centre of Pressure (h^*)

$$\begin{aligned}\text{Pressure force on the strip, } dF &= \rho g h dA \\ &= \rho g y \sin \theta dA\end{aligned}$$

$$\begin{aligned}\text{Moment of the force, } dF, \text{ about axis } O-O \\ &= dF \times y = \rho g y \sin \theta dA \times y = \rho g \sin \theta y^2 dA\end{aligned}$$

$$\begin{aligned}\text{Sum of moments of all such forces about } O-O \\ &= \int \rho g \sin \theta y^2 dA = \rho g \sin \theta \int y^2 dA\end{aligned}$$

$$\text{But} \quad \int y^2 dA = \text{M.O.I. of the surface about } O-O = I_0$$

$$\therefore \text{ Sum of moments of all forces about } O-O = \rho g \sin \theta I_0 \quad \text{eq (vii)}$$

$$\begin{aligned}\text{Moment of the total force, } F, \text{ about } O-O \text{ is also given by} \\ &= F \times y^* \quad \text{eq (viii)}\end{aligned}$$

where y^* = Distance of centre of pressure from $O-O$.

Equating the two values given by equations (vii) and (viii)

$$F \times y^* = \rho g \sin \theta I_0$$

$$\text{or} \quad y^* = \frac{\rho g \sin \theta I_0}{F} \quad \text{eq (ix)}$$

$$\text{Now} \quad y^* = \frac{h^*}{\sin \theta}, \quad F = \rho g A \bar{h}$$

and I_0 by the theorem of parallel axis = $I_G + A \bar{y}^2$.

Substituting these values in equation (3.9), we get

$$\frac{h^*}{\sin \theta} = \frac{\rho g \sin \theta}{\rho g A \bar{h}} [I_G + A \bar{y}^2]$$

$$\therefore h^* = \frac{\sin^2 \theta}{A \bar{h}} [I_G + A \bar{y}^2]$$

But $\frac{\bar{h}}{\bar{y}} = \sin \theta$ or $\bar{y} = \frac{\bar{h}}{\sin \theta}$

$$\therefore h^* = \frac{\sin^2 \theta}{A \bar{h}} \left[I_G + A \times \frac{\bar{h}^2}{\sin^2 \theta} \right]$$

or
$$h^* = \frac{I_G \sin^2 \theta}{A \bar{h}} + \bar{h} \quad \text{eq (x)}$$

If θ is 90° , equation (x) above becomes same as Eq (v) which is applicable to vertically plane submerged surfaces.

Worked Example 1:

A rectangular plane surface 2 m wide and 3 m deep lies in water in such a way that its plane makes an angle of 30 degrees with the free water surface. Determine the total pressure and position of centre of pressure when the upper edge is 1.5 m below the free water surface.

Solution. Given :

Width of plane surface, $b = 2 \text{ m}$

Depth, $d = 3 \text{ m}$

Angle, $\theta = 30^\circ$

Distance of upper edge from free water surface = 1.5 m

(i) Total pressure force is given by equation (3.6) as

$$F = \rho g A \bar{h}$$

where $\rho = 1000 \text{ kg/m}^3$

$$A = b \times d = 3 \times 2 = 6 \text{ m}^2$$

$$\begin{aligned} \therefore \bar{h} &= \text{Depth of C.G. from free water surface} \\ &= 1.5 + 1.5 \sin 30^\circ \end{aligned}$$

$$= 1.5 + 1.5 \times \frac{1}{2} = 2.25 \text{ m}$$

$$\therefore F = 1000 \times 9.81 \times 6 \times 2.25 = \mathbf{132435 \text{ N. Ans.}}$$

(ii) Centre of pressure (h^*)

Using equation (3.10), we have

$$h^* = \frac{I_G \sin^2 \theta}{A \bar{h}} + \bar{h}, \quad \text{where } I_G = \frac{bd^3}{12} = \frac{2 \times 3^3}{12} = 4.5 \text{ m}^4$$

$$\begin{aligned} \therefore h^* &= \frac{4.5 \times \sin^2 30^\circ}{6 \times 2.25} + 2.25 = \frac{4.5 \times \frac{1}{4}}{6 \times 2.25} + 2.25 \\ &= 0.0833 + 2.25 = \mathbf{2.3333 \text{ m. Ans.}} \end{aligned}$$

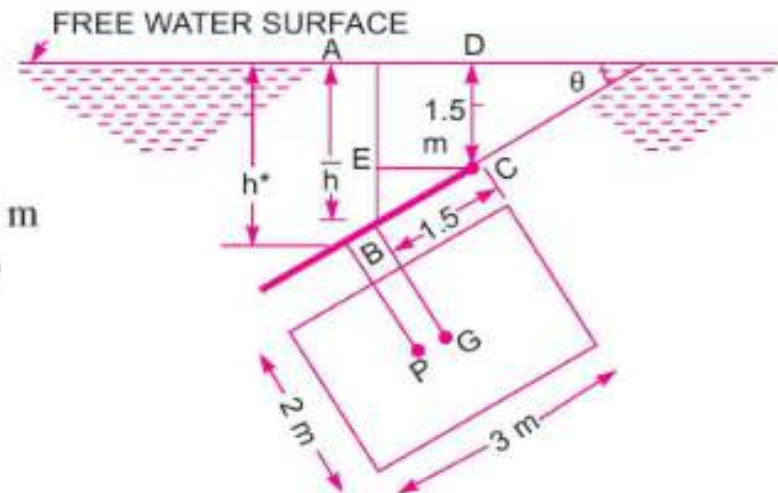


Fig. 3.19

$$\{ \because \bar{h} = AE + EB = 1.5 + BC \sin 30^\circ = 1.5 + 1.5 \sin 30^\circ \}$$

Worked Example 2:

A circular plate 3.0 m diameter is immersed in water in such a way that its greatest and least depth below the free surface are 4 m and 1.5 m respectively. Determine the total pressure on one face of the plate and position of the centre of pressure.

Solution. Given :

Dia. of plate, $d = 3.0 \text{ m}$

$\therefore$ Area, $A = \frac{\pi}{4} d^2 = \frac{\pi}{4} (3.0)^2 = 7.0685 \text{ m}^2$

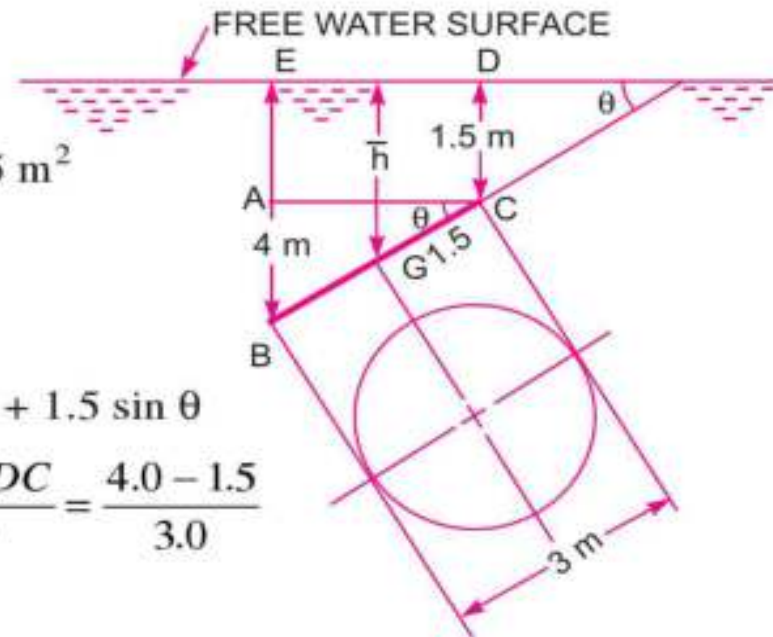
Distance $DC = 1.5 \text{ m}$, $BE = 4 \text{ m}$

Distance of C.G. from free surface

$$= \bar{h} = CD + GC \sin \theta = 1.5 + 1.5 \sin \theta$$

But
$$\sin \theta = \frac{AB}{BC} = \frac{BE - AE}{BC} = \frac{4.0 - DC}{3.0} = \frac{4.0 - 1.5}{3.0}$$
$$= \frac{2.5}{3.0} = 0.8333$$

$$\bar{h} = 1.5 + 1.5 \times 0.8333 = 1.5 + 1.249 = 2.749 \text{ m}$$



(i) **Total pressure (F)**

$$F = \rho g A \bar{h}$$
$$= 1000 \times 9.81 \times 7.0685 \times 2.749 = \mathbf{190621 \text{ N. Ans.}}$$

(ii) Centre of pressure (h^*)

Using equation (3.10), we have $h^* = \frac{I_G \sin^2 \theta}{A \bar{h}} + \bar{h}$

where $I_G = \frac{\pi}{64} d^4 = \frac{\pi}{64} (3)^4 = 3.976 \text{ m}^4$

$$\begin{aligned} h^* &= \frac{3.976 \times (.8333) \times .8333}{7.0685 \times 2.749} + 2.749 = 0.1420 + 2.749 \\ &= \mathbf{2.891 \text{ m. Ans.}} \end{aligned}$$

Worked Example 3:

A rectangular gate 5 m x 2 m is hinged at its base and inclined at 60° to the horizontal as shown in Fig. 3.22. To keep the gate in a stable position, a counter weight of 5000 kgf is attached at the upper end of the gate as shown in figure 3.22. Find the depth of water at which the gate begins to fall. Neglect the weight of the gate and friction at the hinge and pulley.

Solution. Given :

Length of gate = 5 m

Width of gate = 2 m

$\theta = 60^\circ$

Weight, $W = 5000 \text{ kgf}$
 $= 5000 \times 9.81 \text{ N}$
 $= 49050 \text{ N} \quad (\because 1 \text{ kgf} = 9.81 \text{ N})$

As the pulley is frictionless, the force acting at $B = 49050 \text{ N}$. First find the total force F acting on the gate AB for a given depth of water.

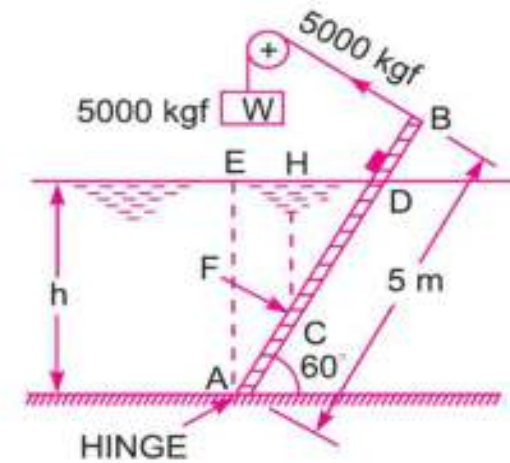


Fig. 3.22

From figure, $AD = \frac{AE}{\sin \theta} = \frac{h}{\sin 60^\circ} = \frac{5}{\sqrt{3}/2} = \frac{2h}{\sqrt{3}}$

$\therefore$ Area of gate immersed in water, $A = AD \times \text{Width} \times \frac{2h}{\sqrt{3}} \times 2 = \frac{4h}{\sqrt{3}} \text{ m}^2$

Also depth of the C.G. of the immersed area $= \bar{h} = \frac{h}{2} = 0.5 h$

$\therefore$ Total force F is given by $F = \rho g A \bar{h} = 1000 \times 9.81 \times \frac{4h}{\sqrt{3}} \times \frac{h}{2} = \frac{19620}{\sqrt{3}} h^2 \text{ N}$

The centre of pressure of the immersed surface, h^* is given by

$$h^* = \frac{I_G \sin^2 \theta}{A \bar{h}} + \bar{h}$$

where $I_G = \text{M.O.I. of the immersed area}$

$$\begin{aligned} &= \frac{b \times (AD)^3}{12} = \frac{2}{12} \times \left(\frac{2h}{\sqrt{3}} \right)^3 \\ &= \frac{16h^3}{12 \times 3 \times \sqrt{3}} = \frac{4h^3}{9 \times \sqrt{3}} \text{ m}^4 \end{aligned}$$

$$\therefore h^* = \frac{4h^3}{9 \times \sqrt{3}} \times \frac{\left(\frac{\sqrt{3}}{2}\right)^2}{\frac{4h}{\sqrt{3}} \times \frac{h}{2}} + \frac{h}{2} = \frac{3h^3}{18h^2} + \frac{h}{2} = \frac{h}{6} + \frac{h}{2} = \frac{h+3h}{6} = \frac{2h}{3}$$

Now in the $\triangle CHD$, $CH = h^* = \frac{2h}{3}$, $\angle CDH = 60^\circ$

$$\therefore \frac{CH}{CD} = \sin 60^\circ$$

$$\therefore CD = \frac{CH}{\sin 60^\circ} = \frac{h^*}{\sin 60^\circ} = \frac{2h}{3 \times \frac{\sqrt{3}}{2}} = \frac{4h}{3 \times \sqrt{3}}$$

$$\therefore AC = AD - CD = \frac{2h}{\sqrt{3}} - \frac{4h}{3\sqrt{3}} = \frac{6h - 4h}{3\sqrt{3}} = \frac{2h}{3\sqrt{3}} \text{ m}$$

Taking the moments about hinge, we get

$$49050 \times 5.0 = F \times AC = \frac{19620}{\sqrt{3}} h^2 \times \frac{2h}{3\sqrt{3}}$$

or $.245250 = \frac{39240 h^3}{3 \times 3}$

$$\therefore h^3 = \frac{9 \times 245250}{39240} = 56.25$$

$$\therefore h = (56.25)^{1/3} = \mathbf{3.83 \text{ m. Ans.}}$$

CURVED SURFACE SUBMERGED IN LIQUID

Consider a curved surface AB, submerged in a static fluid as shown in Fig. 3.27. Let dA be the area of a small strip at a depth h from water surface.

Pressure intensity on the area $dA = \rho gh$ and pressure force

$$\text{and pressure force } dF = p \times \text{Area} = \rho gh \times dA \quad \text{eq (i)}$$

The force dF acts normal to the surface. Hence, the total pressure force on the curved surface should be given by:

$$F = \int \rho gh dA \quad \text{eq (ii)}$$

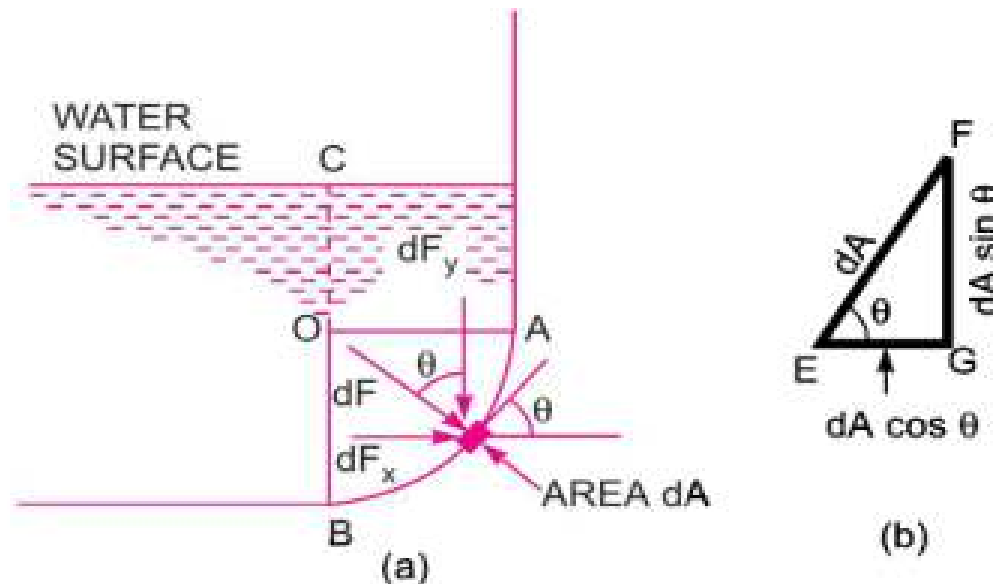


Fig. 3.27

Resolving the force dF into two components dF_x and dF_y in x and y directions respectively, Total force on the curved surface is given by:

$$F = \sqrt{F_x^2 + F_y^2} \quad \text{eq (iii)}$$

Inclination of resultant with the horizontal is $\tan \phi = \frac{F_y}{F_x}$ eq (iv)

Resolving the force dF given by equation (3.11) in x and y directions :

$$dF_x = dF \sin \theta = \rho g h dA \sin \theta$$

and

$$dF_y = dF \cos \theta = \rho g h dA \cos \theta$$

Total forces in the x and y direction are :

$$F_x = \int dF_x = \int \rho g h dA \sin \theta = \rho g \int h dA \sin \theta \quad \text{eq (v)}$$

and

$$F_y = \int dF_y = \int \rho g h dA \cos \theta = \rho g \int h dA \cos \theta \quad \text{eq (vi)}$$

Fig. 3.27 (b) shows the enlarged area dA . From this figure, *i.e.*, $\triangle EFG$,

$$EF = dA$$

$$FG = dA \sin \theta$$

$$EG = dA \cos \theta$$

Thus; in equation (v), $dA \sin \theta = FG$ = Vertical projection of the area dA and hence the expression $\rho g \int h dA \sin \theta$ represents the total pressure force on the projected area of the curved surface on the *vertical plane*.

Thus: F_x = Total pressure force on the projected area of the curved surface on vertical plane.

Similarly, $dA \cos \theta = EG$ = horizontal projection of dA and hence $h dA \cos \theta$ is the volume of the liquid contained in the elementary area dA up to free surface of the liquid.

Thus; $\int h dA \cos \theta$ is the total volume contained between the curved surface extended up to free surface.

Hence $\rho g \int h dA \cos \theta$ is the total weight supported by the curved surface.
Thus

$$F_y = \rho g \int h dA \cos \theta$$

= weight of liquid supported by the curved surface up to free liquid surface

Worked Example 1:

Compute the horizontal and vertical components of the total force acting on a curved surface AB, which is in the form of a quadrant of a circle of radius 2 m as shown in fig. 3.29. Take the width of the gate as unity.

Solution. Given :

Width of gate = 1.0 m

Radius of the gate = 2.0 m

∴ Distance $AO = OB = 2$ m

Horizontal force, F_x exerted by water on gate is given by equation (3.17) as

$F_x =$ Total pressure force on the projected area of curved surface AB on vertical plane

= Total pressure force on OB

{projected area of curved surface on vertical plane = $OB \times 1$ }

$$= \rho g A \bar{h}$$

$$= 1000 \times 9.81 \times 2 \times 1 \times \left(1.5 + \frac{2}{2}\right)$$

{ ∵ Area of OB = $A = BO \times 1 = 2 \times 1 = 2$,

$\bar{h}$ = Depth of C.G. of OB from free surface = $1.5 + \frac{2}{2}$ }

$$F_x = 9.81 \times 2000 \times 2.5 = 49050 \text{ N. Ans.}$$

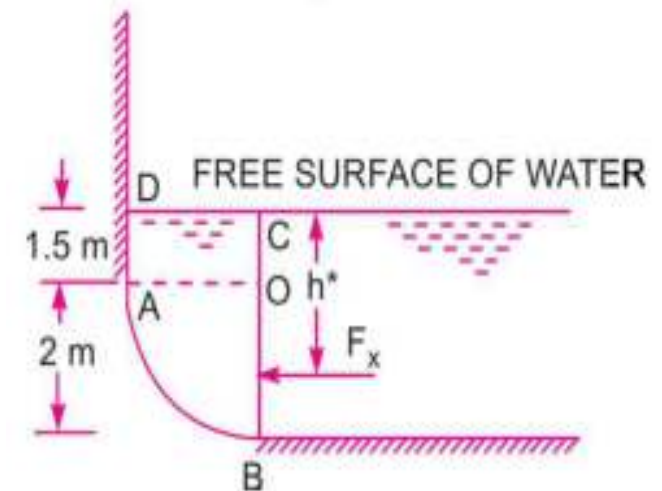


Fig. 3.29

The point of application of F_x is given by $h^* = \frac{I_G}{Ah} + \bar{h}$

where $I_G = \text{M.O.I. of } OB \text{ about its C.G.} = \frac{bd^3}{12} = \frac{1 \times 2^3}{12} = \frac{2}{3} \text{ m}^4$

$$\begin{aligned}\therefore h^* &= \frac{\frac{2}{3}}{2 \times 2.5} + 2.5 = \frac{1}{7.5} + 2.5 \text{ m} \\ &= 0.1333 + 2.5 = 2.633 \text{ m from free surface.}\end{aligned}$$

Vertical force, F_y , exerted by water is given by equation (3.18)

$$\begin{aligned}F_y &= \text{Weight of water supported by } AB \text{ upto free surface} \\ &= \text{Weight of portion } DABOC \\ &= \text{Weight of } DAOC + \text{Weight of water } AOB \\ &= \rho g [\text{Volume of } DAOC + \text{Volume of } AOB] \\ &= 1000 \times 9.81 \left[AD \times AO \times 1 + \frac{\pi}{4} (AO)^2 \times 1 \right] \\ &= 1000 \times 9.81 \left[1.5 \times 2.0 \times 1 + \frac{\pi}{4} \times 2^2 \times 1 \right] \\ &= 1000 \times 9.81 [3.0 + \pi] \text{ N} = \mathbf{60249.1 \text{ N. Ans.}}\end{aligned}$$

Worked Example 2:

Find the magnitude and direction of the resultant water pressure acting on a curved face of a dam which is shaped according to the relation $y = \frac{x^2}{9}$ as shown in Fig. 3.37. The height of the water retained by the dam is 10m. Consider the width of the dam as unity.

Solution. Equation of curve AB is

$$y = \frac{x^2}{9} \quad \text{or} \quad x^2 = 9y$$

$$\therefore x = \sqrt{9y} = 3\sqrt{y}$$

$$\text{Height of water, } h = 10 \text{ m}$$

$$\text{Width, } b = 1 \text{ m}$$

The horizontal component, F_x is given by

$$\begin{aligned} F_x &= \text{Pressure due to water on the curved area projected on vertical plane} \\ &= \text{Pressure on area } BC \\ &= \rho g A \bar{h} \end{aligned}$$

$$\text{where } A = BC \times 1 = 10 \times 1 \text{ m}^2, \bar{h} = \frac{1}{2} \times 10 = 5 \text{ m}$$

$$F_x = 1000 \times 9.81 \times 10 \times 5 = 490500 \text{ N}$$

Vertical component, F_y is given by

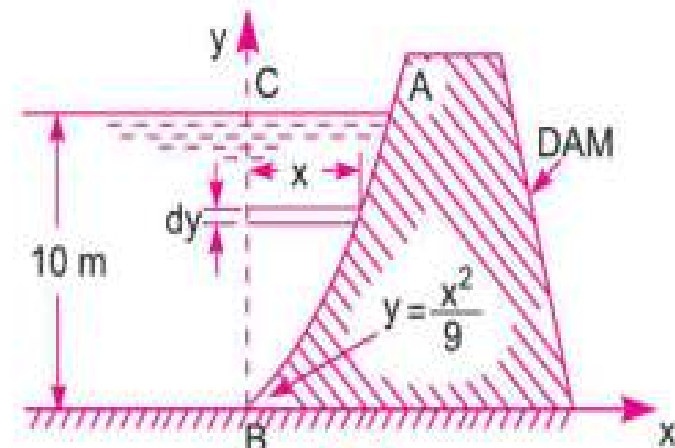


Fig. 3.37

F_y = Weight of water supported by the curve AB

= Weight of water in the portion ABC

= ρg [Area of ABC] $\times$ Width of dam

$$= \rho g \left[\int_0^{10} x \times dy \right] \times 1.0 \quad \left\{ \text{Area of strip} = xdy \quad \therefore \text{Area } ABC = \int_0^{10} xdy \right\}$$

$$= 1000 \times 9.81 \times \int_0^{10} 3\sqrt{y} \, dy \quad \{ \because x = 3\sqrt{y} \}$$

$$= 29430 \left[\frac{y^{3/2}}{3/2} \right]_0^{10} = 29430 \times \frac{2}{3} \left[y^{3/2} \right]_0^{10} = 19620 [10^{3/2}]$$

$$= 19620 \times 31.622 = 620439 \text{ N}$$

$\therefore$ Resultant water pressure on dam

$$\begin{aligned} F &= \sqrt{F_x^2 + F_y^2} = \sqrt{(490500)^2 + (620439)^2} \\ &= 790907 \text{ N} = \mathbf{790.907 \text{ kN. Ans.}} \end{aligned}$$

Direction of the resultant is given by

$$\tan \theta = \frac{F_y}{F_x} = \frac{620439}{490500} = 1.265$$

$$\therefore \theta = \mathbf{51^\circ 40'}. \text{ Ans.}$$

Worked Example 3:

A dam has a parabolic shape $y = y_0 \left(\frac{x}{x_0} \right)^2$ as shown in Fig. 3.38 below having $x_0 = 6 \text{ m}$ and $y_0 = 9 \text{ m}$. The fluid is water with density $= 1000 \text{ kg/m}^3$. Compute the horizontal, vertical and the resultant thrust exerted by water per metre length of the dam.

Solution. Given :

Equation of the curve OA is

$$y = y_0 \left(\frac{x}{x_0} \right)^2 = 9 \left(\frac{x}{6} \right)^2 = 9 \times \frac{x^2}{36} = \frac{x^2}{4}$$

or $x^2 = 4y$

$\therefore x = \sqrt{4y} = 2y^{1/2}$

Width of dam, $b = 1 \text{ m}$.

(i) **Horizontal thrust exerted by water**

F_x = Force exerted by water on vertical surface OB, i.e., the surface obtained by projecting the curved surface on vertical plane

$$= \rho g A \bar{h}$$

$$= 1000 \times 9.81 \times (9 \times 1) \times \frac{9}{2} = 397305 \text{ N. Ans.}$$

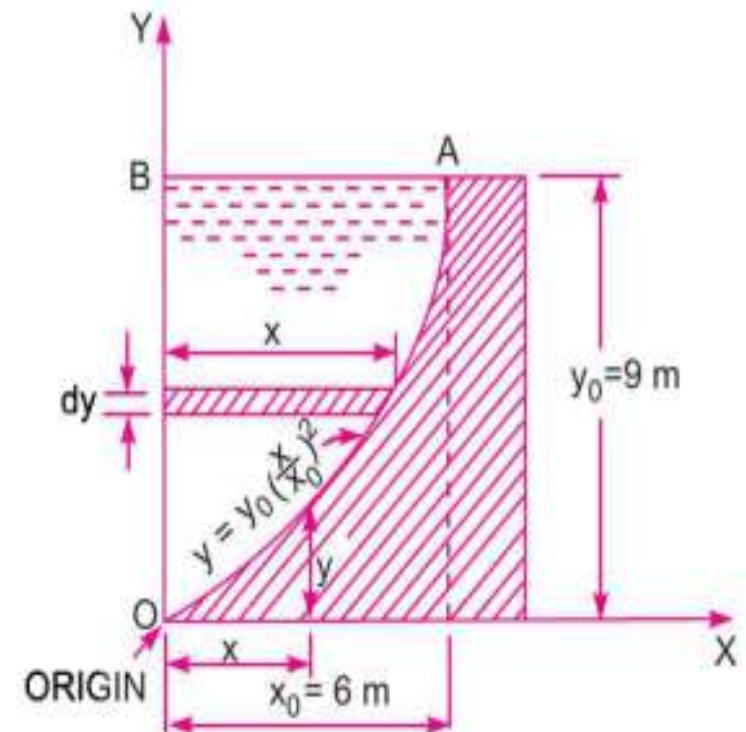


Fig. 3.38

(ii) Vertical thrust exerted by water

F_y = Weight of water supported by curved surface OA upto free surface of water

= Weight of water in the portion ABO

= $\rho g \times \text{Area of } OAB \times \text{Width of dam}$

$$= 1000 \times 9.81 \times \left[\int_0^9 x \times dy \right] \times 1.0$$

$$= 1000 \times 9.81 \times \left[\int_0^9 2y^{1/2} \times dy \right] \times 1.0 \quad (\because x = 2y^{1/2})$$

$$= 19620 \times \left[\frac{y^{3/2}}{(3/2)} \right]_0^9 = 19620 \times \frac{2}{3} [9^{3/2}]$$

$$= 19620 \times \frac{2}{3} \times 27 = \mathbf{353160 \text{ N. Ans.}}$$

(iii) Resultant thrust exerted by water

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{397305 + 353160} = \mathbf{531574 \text{ N. Ans.}}$$

Direction of resultant is given by

$$\tan \theta = \frac{F_y}{F_x} = \frac{353160}{397305} = 0.888$$

$$\theta = \tan^{-1} 0.888 = \mathbf{41.63^\circ. \text{ Ans.}}$$

Worked Example 4:

A cylinder 3 m in dia and 4 m long retains water on one side. The cylinder is supported as shown in Fig. 3.39. Determine the horizontal reaction at A and the vertical reaction at B. The cylinder weighs 196.2 kN. Ignore friction.

Solution. Given :

Dia. of cylinder = 3 m

Length of cylinder = 4 m

Weight of cylinder, $W = 196.2 \text{ kN} = 196200 \text{ N}$

Horizontal force exerted by water

$$\begin{aligned} F_x &= \text{Force on vertical area } BOC \\ &= \rho g A \bar{h} \end{aligned}$$

$$\text{here } A = BOC \times l = 3 \times 4 = 12 \text{ m}^2, \bar{h} = \frac{1}{2} \times 3 = 1.5 \text{ m}$$

$$F_x = 1000 \times 9.81 \times 12 \times 1.5 = 176580 \text{ N}$$

The vertical force exerted by water

$$\begin{aligned} F_y &= \text{Weight of water enclosed in } BDCOB \\ &= \rho g \times \left(\frac{\pi}{2} R^2 \right) \times l = 1000 \times 9.81 \times \frac{\pi}{2} \times (1.5)^2 \times 4 = 138684 \text{ N} \end{aligned}$$

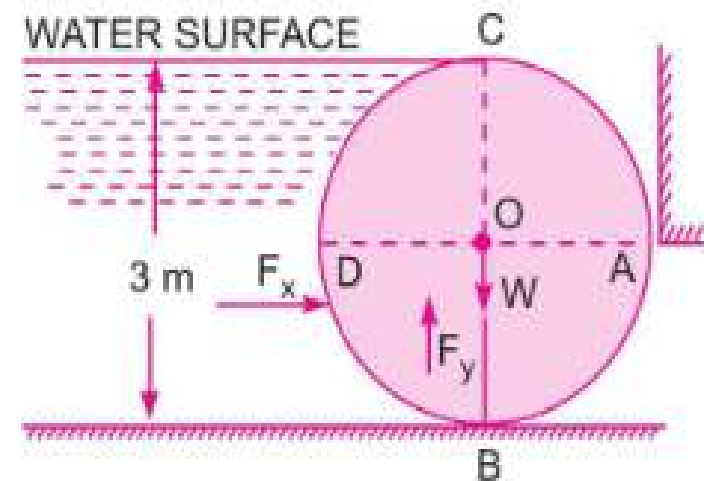


Fig. 3.39

Force F_y is acting in the upward direction.

For the equilibrium of cylinder

Horizontal reaction at $A = F_x = 176580 \text{ N}$

Vertical reaction at $B = \text{Weight of cylinder} - F_y$
 $= 196200 - 138684 = \mathbf{57516 \text{ N. Ans.}}$

DAMS & DAM FAILURES

A dam is a barrier constructed to hold back water and raise its level, forming a reservoir used for various purposes. The reservoirs created by dams can be used for the following purposes:

Flood suppression, irrigation, human consumption through water supply system, industrial use, aquaculture, navigation, recreation and power generation. Hydropower is often used in conjunction with dams to generate electricity.

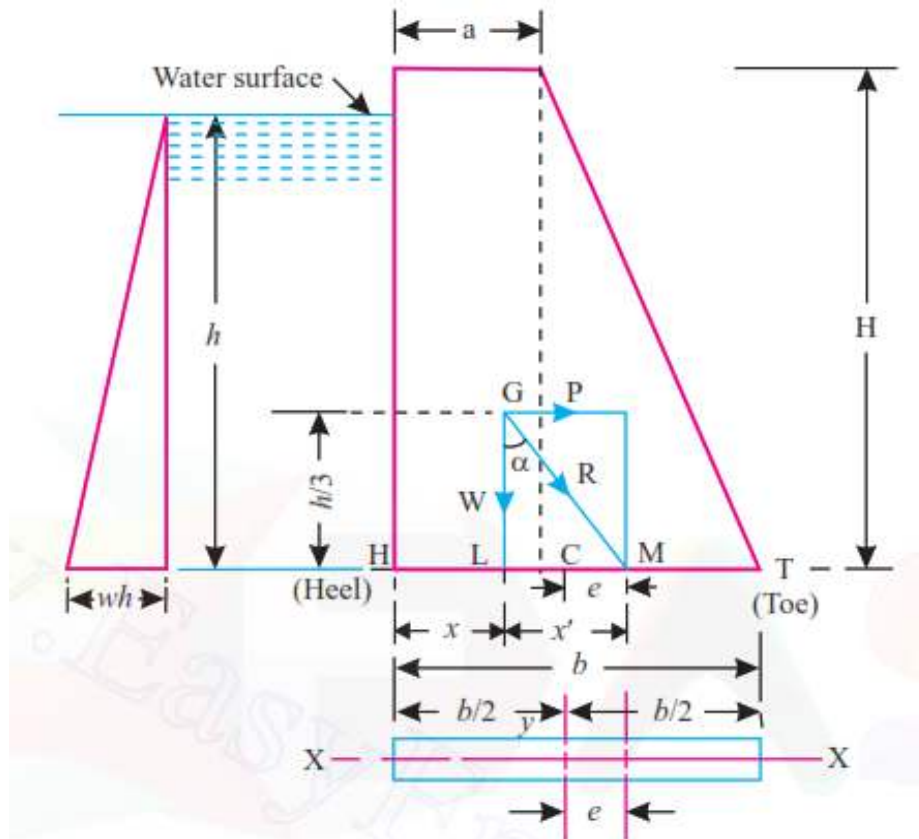
Dams may be used to meet more than one of the above activities and those dams are referred to as multipurpose dams

Possibilities of Dam Failure

The following are the possibilities of dam failure:

- i. Failure due to sliding along its base
- ii. Failure due to tension or compression
- iii. Failure due to shear at the weakest section
- iv. Failure due to overturning

Consider a trapezoidal shape dam illustrated below



a = Top width of the dam,
 b = Base width of the dam,
 H = Height of the dam, and
 h = Height of water column.

Consider 1 m length of the dam.

Weight of masonry = Area $\times$ length $\times$ density of masonry

$$\therefore W = \left(\frac{a+b}{2} \right) \times H \times 1 \times \text{density of masonry}$$

Let c.g of the section be x distance from the vertical face. Dividing the trapezium into rectangle and triangle and taking moments about the vertical face,

$$\begin{aligned}
 & a \times h \times \frac{a}{2} + \frac{1}{2} (b-a) \times h \left[a + \left(\frac{b-a}{3} \right) \right] \\
 &= \left[a \times h + \left(\frac{b-a}{2} \right) \times h \right] \bar{x} \\
 \bar{x} &= \frac{a \times h \times \frac{a}{2} + \frac{1}{2} (b-a) \times h \left[a + \frac{b-a}{3} \right]}{a \times h + \left(\frac{b-a}{2} \right) \times h} \\
 \bar{x} &= \frac{a^2 + ab + b^2}{3(a+b)}
 \end{aligned}$$

Total water pressure (P) = Area $\times$ average pressure

$$P = (l \times h) \times \frac{wh}{2} = \frac{wh^2}{2}$$

Stresses at the base:

$$\text{Direct stress, } \sigma_d = \frac{\text{Weight of masonry}}{\text{Area at the base}} = \frac{W}{b \times 1} = \frac{W}{b} \text{ (compressive)}$$

$$\text{Bending stress, } \sigma_b = \pm \frac{My}{I} = \frac{(W.e) \times b/2}{1/12 \times l \times b^3}$$

$$= \pm \frac{6 W.e}{b^2} \text{ (-ve sign stands for tensile stress here)}$$

Maximum intensity of stress,

$$\begin{aligned} \sigma_{\max} &= \sigma_d + \sigma_b \\ &= +\frac{W}{b} + \frac{6We}{b^2} = +\frac{W}{b} \left(1 + \frac{6e}{b} \right) \text{ (Compressive)} \end{aligned}$$

Minimum intensity of stress,

$$\sigma_{\min} = \sigma_d + (-\sigma_b) = \frac{W}{b} - \frac{6We}{b^2} = \frac{W}{b} \left(1 - \frac{6e}{b} \right) \text{ (Compressive)}$$

(i) Failure due to sliding along its base

Sliding may occur due to horizontal water pressure (P) on the walls of the dam. This pressure must be resisted by the dam's foundation to ensure an equilibrium state.

The dam will be stable against sliding if the frictional resistance is more than the sliding force P .

Now, frictional force, $F = \mu W$

Sliding or driving force = P

Factor of safety against sliding = $\frac{\mu W}{P}$

(ii) Failure due to tension or compression

The dam will be stable if no tensile stress across the cross-section is produced. It means σ_d should be more than or equal to σ_c . In certain cases where tension cannot be avoided it should not increase more than 0.4 N/mm²

$$\sigma_d \geq \sigma_c$$
$$\frac{W}{b} \geq \frac{6W \cdot e}{b^2} \quad \text{or} \quad 1 \geq \frac{6 \times e}{b} \quad \text{or} \quad e \leq \frac{b}{6}$$

$$e = \frac{b}{6}$$

$$HM = x + x' = \frac{b}{2} + e = \frac{b}{2} + \frac{b}{6} = \frac{2b}{3}$$

(iii) Failure due to shear at the weakest section

If, A' = The least cross-sectional area of the dam at any section, and

$\sigma_{s(\max)}$ = Maximum safe shear stress of the dam material,

Then, resistance against shear = $\sigma_{s(\max)} \times A'$

Factor of safety against shear = $\frac{\sigma_{s(\max)} \times A'}{P'}$ (It must be *greater than unity*)

where, P' = Total liquid pressure due to water column *above the section*.

(iv) Failure due to overturning

Referring to Fig. 3.56 we find that water pressure tends to overturn the wall about the toe T whereas W tends to counteract the turning effect.

Taking moments of P and W about toe T , we get overturning moment = $P \times h/3$.

and resisting moment = $W(b - x)$

Factor of safety against overturning = $\frac{W(b - x)}{P \times (h/3)}$

= $\frac{3W(b - x)}{Ph}$ (It must be *greater than unity*)

Worked Example 1:

A concrete dam of trapezoidal section having water on vertical face is 12 m high. The base of the dam is 8 m wide and top 2 m wide. Find the resultant thrust on the base per m-length of the dam, and the point where it intersects the base. Take the specific weight of masonry as 240 kN/m^3 and water level coinciding with the top of the dam.

Solution. Refer to Fig. 3.57.

Top width = 2 m, base width = 8 m, height = 12 m

Consider 1 m length of dam.

Weight of masonry,

$$W = \left(\frac{8+2}{2} \right) \times 12 \times 1 \times 24 = 1440 \text{ kN}$$

Water pressure,

$$P = \frac{wh^2}{2} = \frac{9.81 \times 12^2}{2} = 706 \text{ kN}$$

$$\text{Resultant thrust, } R = \sqrt{P^2 + W^2}$$

$$= \sqrt{(706)^2 + (1440)^2} = 1604 \text{ kN (Ans.)}$$

The point, where the resultant thrust cuts the base:

Let x metres be the distance of c.g. from the vertical face. Dividing the trapezium into rectangle and triangle and taking moments about the vertical face, we have:

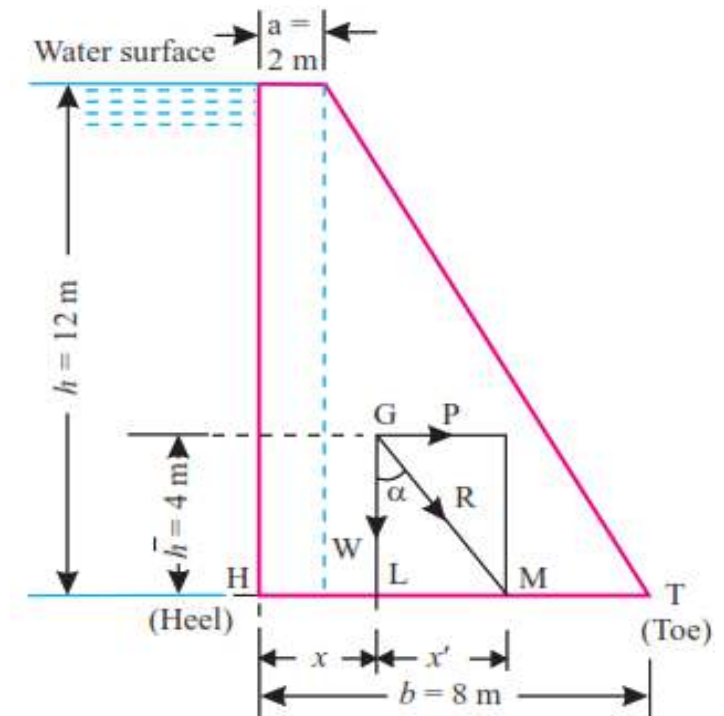


Fig. 3.57

$$\begin{aligned} & 12 \times 2 \times \frac{2}{2} + \frac{1}{2} \times 12 \times 6 \left(2 + \frac{6}{3} \right) \\ &= (12 \times 2 + \frac{1}{2} \times 12 \times 6) \times x \\ &= 24 + 36 \times 4 = 60x \end{aligned}$$

or,
$$x = \frac{24 + 36 \times 4}{60} = 2.8 \text{ m}$$

The value of 'x' can also be found by using the relation:

$$x = \frac{a^2 + ab + b^2}{3(a + b)} = \frac{2^2 + 2 \times 8 + 8^2}{3(2 + 8)} = 2.8 \text{ m}$$

From Fig. 3.57,
$$\tan \alpha = \frac{P}{W} = \frac{706}{1440}$$

Also,
$$\tan \alpha = \frac{x'}{4} = \frac{706}{1440} \text{ or } x' = \frac{706 \times 4}{1440} = 1.96 \text{ m}$$

Now, the point where the resultant thrust cuts the base

$$= x + x' = 2.8 + 1.96 = 4.76 \text{ m from H (heel) (Ans.)}$$

Worked Example 2:

A masonry dam trapezoidal in cross section is 4 m wide at the top, 8 m wide at the base and 10 m high. It retains water level with top against a vertical face. Obtain stress distribution at the base if specific gravity of masonry is 2.5.

Solution. Refer to Fig. 3.58.

Top width = 4 m; base width = 8 m; height = 10 m

Density of masonry = $2.5 \times 9.81 = 24.5 \text{ kN/m}^3$

Consider 1 m length of the dam.

Weight of masonry acting through c.g.,

$$W = \left(\frac{8+4}{2} \right) \times 10 \times 1 \times 24.5 = 1470 \text{ kN}$$

$$\text{Water pressure, } P = \frac{wh^2}{2} = \frac{9.81 \times 10^2}{2} = 490.5 \text{ kN}$$

Let x be the distance of c.g. from the vertical face.

Dividing the trapezium into triangle and rectangle and taking moments about vertical face, we have

$$10 \times 4 \times \frac{4}{2} + \frac{1}{2} \times 10 \times 4 \left(4 + \frac{4}{3} \right)$$

$$= \left(10 \times 4 + \frac{1}{2} \times 10 \times 4 \right) \times x$$

$$\text{or, } 80 + 20 \times 5.333 = 60x$$

$$x = \frac{80 + 20 \times 5.33}{60} = 3.11 \text{ m}$$

From Fig. 3.58.,

$$\tan \alpha = \frac{P}{W} = \frac{490.5}{1470}$$

$$\text{Also, } \tan \alpha = \frac{x'}{10/3}$$

$$\therefore \frac{x'}{10/3} = \frac{490.5}{1470} = \frac{1}{3}$$

$$\text{or, } x' = \frac{1}{3} \times \frac{10}{3} = \frac{10}{9} = 1.11 \text{ m}$$

$$\text{Now, } x + x' = 3.11 + 1.11 = 4.22 \text{ m}$$

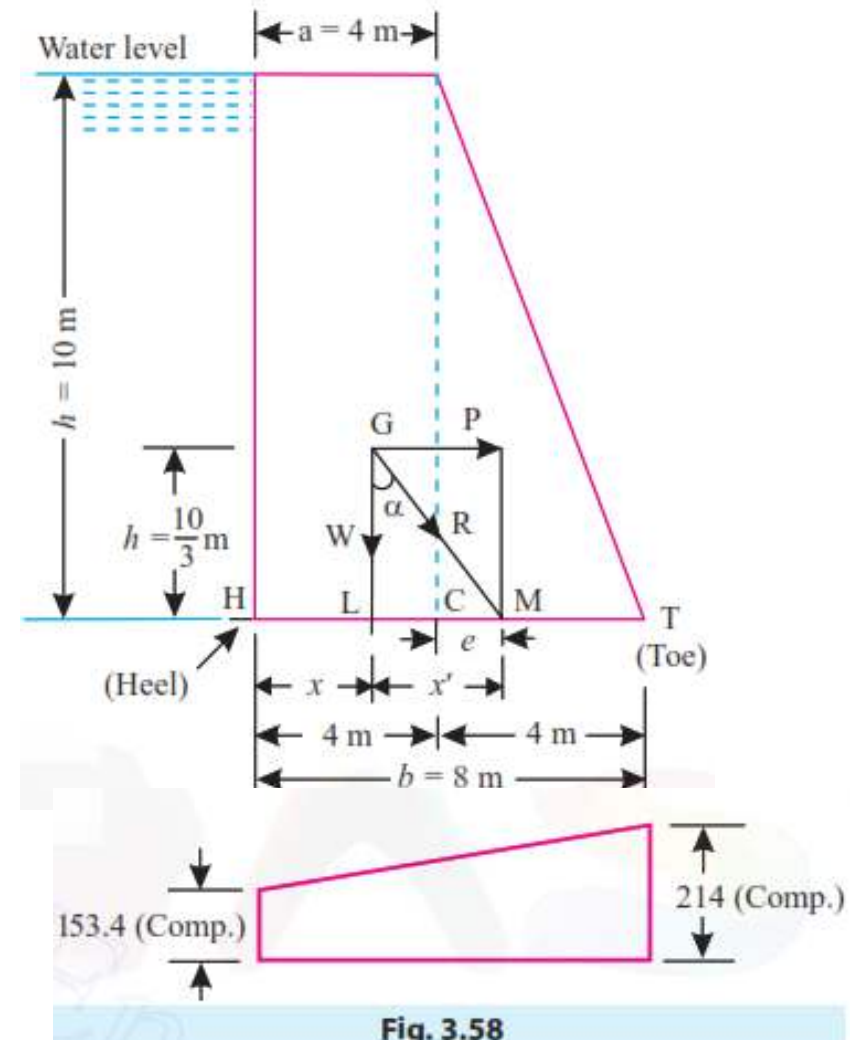


Fig. 3.58

Bending stress,

$$\sigma_b = \frac{M}{I} y = \frac{(W.e).y}{I}$$

$$= \frac{1470 \times 0.22 \times 8/2}{\frac{1 \times 8^3}{12}} = \frac{1470 \times 0.22 \times 4 \times 12}{8^3} = \pm 30.3 \text{ kN/m}^2$$

Direct stress,

$$\sigma_d = \frac{\text{Weight of masonry}}{\text{Area of base}} = \frac{1470}{8 \times 1} = 183.7 \text{ kN/m}^2$$

$$\sigma_{\max} = \sigma_d + \sigma_b = 183.7 + 30.3 = \mathbf{214 \text{ kN/m}^2 \text{ (comp.) (Ans.)}}$$

$$\sigma_{\min} = \sigma_d - \sigma_b = 183.7 - 30.3 = \mathbf{153.4 \text{ kN/m}^2 \text{ (comp.) (Ans.)}}$$